DATA ANALYTICS - PHASE 1 REPORT

Name: Veni Prasanna S **Project:** PulseIQ Analytics
Phase 1: Tasks 1- 5 **Track:** Data Analytics
GitHub repository: https://github.com/VENIPRASANNA/growmar_pulseiq

1.Project Overview:
PulseIQ is a workforce analytics project intended to organize workforce-related data, validate data quality, define governed metrics, and present insights through a dashboard. This report describes the implement evidence available so far and identifies items that should be verified before final submission.

2.Tasks:
Task 1 — Data Warehouse
Task 2 — ETL Pipeline
Task 3 — Data Quality
Task 4 — Metric Layer
Task 5 — Dashboard

3.ETS results:
- fact_anomaly: 405 input rows, 405 passed, 0 quarantined.
- fact_attrition_risk: 430 input rows, 425 passed, 5 quarantined.
- fact_forecast: 72 input rows, 72 passed, 0 quarantined.
- fact_task: 28,868 input rows, 28,848 passed, 20 quarantined.

4.Metric Validation:
The validator output confirms 28 metric views and 28 catalog definitions, and reports 15 passed checks with no failed checks. It also explicitly marks nine thresholds as requiring review.
M26 is unavailable because the scored population is incomplete. M28 has zero scored rows because actual headcount months do not match the forecast months. These constraints should be described clearly so readers do not mistake missing results for a zero business outcome.

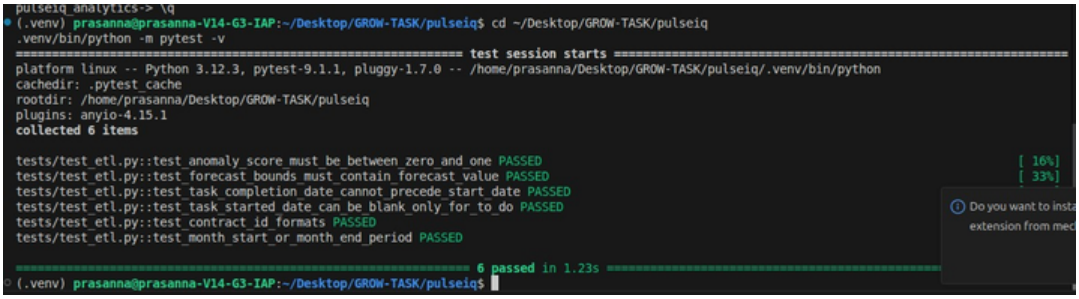
5.Dashboard Evidences:
- Figure 1 - Main dashboard: title, KPI cards, and available charts.



- Figure 2 - Workforce Details: department, location-tier, and work-mode visualizations.



- Figure 3 - Department headcount share, if the chart loads correctly.



6.Challenges, Limitations, and Follow-up:

- Nine metric thresholds need business review.
- M26 cannot be evaluated with the incomplete scored population.
- M28 cannot be scored without matching actual and forecast periods.
- The anomaly source-reference note is not verified because the referenced feature source datasets are absent from the raw schema.
- Unexpected columns are recorded in the ETL summary and should be reviewed against the schema contracts.

7.Learning Outcomes:

• Using automated tests and validation scripts to check implementation behavior.
• Designing and checking governed metric definitions.
• Presenting analytical results in a dashboard.

8.Phase 2 Plan:

• Review and approve metric privacy thresholds with the relevant stakeholders.
• Improve or document missing source data needed for anomaly scoring.
• Align actual and forecast periods before reporting forecast accuracy.
• Review unexpected columns and update data contracts where appropriate.

7.Conclusion:

Phase 1 has implementation evidence for the ETL pipeline, a governed metric layer, automated tests, and dashboard files. The ETL log records a successful run, the metric validator reports 15 passed and 0 failed, and the repository is tagged phase-1 at commit f0048fe. Some thresholds and data-availability limitations remain and should be clearly documented.